

Ahmed Hassen, PhD

Robotics • Mechatronics • Computer Vision • Applied ML Scientist

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SUMMARY

Robotics and Computer Vision Engineer with a Mechanical Design, Manufacturing and Automation foundation, specializing in perception and control systems validated on real hardware — including a markerless 6D pose estimation pipeline that cut translation error $6.75\times$ and rotation error $14.31\times$ over standard PnP baselines at sub-100ms GPU inference. Hands-on experience across CAD modeling, 6D pose estimation, semantic segmentation, Vision-Language-Action (VLA) Models, imitation learning, and reinforcement learning-based control, taking systems from formulation through sim-to-real validation. Experienced authoring and publishing peer-reviewed articles in top journals and conferences, with research spanning perception, segmentation, and control.

TECHNICAL SKILLS

- **ML / Computer Vision:** PyTorch, Vision-Language Models (VLMs), Vision-Language-Action (VLA) Models, Imitation Learning (Diffusion Policy, ACT), OpenCV, Vision Transformers (ViT), CNNs, MLPs, 6D pose estimation
- **Robotics:** ROS/ROS2, MoveIt, Vision-Language-Action (VLA) Control, Simulated Teleoperation & Demonstration Collection, impedance & admittance control, force/torque control, sensor integration
- **Reinforcement Learning & Simulation:** MuJoCo, Isaac Sim, Gymnasium, Stable-Baselines3, SAC/PPO/TD3/DDPG, sim-to-real transfer
- **ML Lifecycle & Inference:** Dataset Curation, End-to-End Model Training, Fine-Tuning, Real-Time Inference Optimization, Latency Profiling, Model Quantization/Pruning, Synthetic-to-Real Data Pipelines
- **Hardware:** UR5 manipulators, 6-axis force/torque sensors, vacuum grippers, RGB cameras, real-time control loops
- **Tools & Infrastructure:** Linux, Docker, Git, ONNX/TensorRT, LaTeX, SolidWorks, Python, C++, MATLAB
- **Professional:** Technical leadership, cross-functional collaboration, rapid prototyping, problem-solving, operational discipline

EXPERIENCE

Robotics Researcher (Ph.D. Candidate)

2024 – Present

Monash University, Melbourne, Australia

- Designed and trained LANet, a lightweight real-time semantic segmentation network for curtain wall frame detection: achieved 84.26% mIoU with 1.92M parameters, running at 262 FPS (GPU) and 19 FPS (CPU) — matching heavier baselines at 1/7th the parameter count to enable on-site edge deployment.
- Built a markerless, monocular 6D pose estimation pipeline for robotic facade assembly, achieving 2.24 mm median in-plane and 11.59 mm median 3D translation error with 4.91° median orientation error across 36 target cases.
- Integrated and benchmarked state-of-the-art architectures (DINOv3 Vision Transformers, YOLO detectors, PIDNet), optimizing the vision system to maintain end-to-end GPU inference under 100ms.
- Engineered a custom neural network operating in SE(3) twist-space for data-driven residual error correction, successfully reducing standard PnP geometric solver errors by 6.7x for translation and 14.3x for rotation.
- Architected an automated pose-error regression pipeline featuring custom rejection-gate logic to generate a high-quality 10,000-sample dataset, alongside an expanded 9,191-image hybrid dataset bridging the sim-to-real gap.
- Developed a safety-aware reinforcement learning framework for variable admittance control, benchmarking different RL algorithms.
- Executed zero-shot sim-to-real transfer of learned policies onto a UR5 testbed via a ROS2 control interface, containerized with Docker, equipped with a vacuum gripper and 6-axis F/T sensor.
- Managed the complete project lifecycle: problem definition, system implementation, experimental validation, and technical documentation.

Assistant Lecturer (Part-time)

July 2024 – Present

Monash University, Melbourne, Australia

- Design and deliver engineering tutorials, examination problems, and technical assessment rubrics, translating complex control theory concepts into accessible frameworks for diverse undergraduate cohorts.
- Collaborate with unit coordinators and fellow teaching associates on instructional planning, course administration, and consistent delivery across multiple tutorial groups.
- Mark designated sections of exams and tests for a unit-wide cohort of 1,200+ undergraduate students, ensuring accurate, timely, and constructive feedback under recurring deadlines.

Technical Leadership & Mentorship

2025 – Present

Monash University, Melbourne, Australia

- Lead technical direction and day-to-day mentoring for 3 Final Year Engineering Capstone students; guide system architecture, algorithm implementation, and experimental design to ensure project delivery.
- Co-supervise onboarding PhD researchers, providing hands-on technical guidance across PyTorch deep learning pipelines, research question formulation, physics simulation modelling, and hardware setup.

EDUCATION

Ph.D., Mechanical Engineering (Robotics & Computer vision) <i>Monash University, Australia</i>	2024 – 2027
Master of Engineering, Aerospace Science and Technology <i>Beihang University, China</i>	2021 – 2023
Bachelor of Engineering, Mechanical Design, Manufacturing and Automation <i>University of Electronic Science and Technology of China (UESTC)</i>	2017 – 2021

SELECTED PUBLICATIONS

1. **Hassen, A. Y.**, Arashpour, M., & Abdi, E. (2026). Lightweight vision-based 6D pose estimation of curtain wall module installation slot. *Expert Systems with Applications*, 133873.
2. **Hassen, A. Y.**, Arashpour, M., & Abdi, E. (2025). Lightweight segmentation model for automated facade installation in high-rise buildings. *Advanced Engineering Informatics*, 65, 103374.
3. **Hassen, A. Y.**, Johns, B., Arashpour, M., & Abdi, E. Facade semantic segmentation for curtain wall installation robotics: A markerless 3D pose estimation. *Robotics and Autonomous Systems* (**under review**).
4. **Hassen, A. Y.**, Arashpour, M., & Abdi, E. Reinforcement Learning-Based Variable Admittance Control for Robotic Curtain Wall Module Installation. *IEEE Transactions on Automation Science and Engineering* (**under review**).
5. Biyogo, N. V., Shi, P., Saeed, A., **Hassen, A. Y.**, & Babansoro, M. (2023). Study on attitude deviation of spacecraft subjected to random disturbance torque based on covariance analysis. *6th International Conference on Aeronautical, Aerospace and Mechanical Engineering (AAME)*.
6. **Hassen, A. Y.**, & Gong, S. (2022). Adaptive-fuzzy-logic based controller for attitude control of solar sail spacecraft. *28th International Symposium on Space Flight Dynamics (ISSFD)*, Beijing, China.

PROJECTS

Teleoperated Demonstration Learning on a Franka Panda

- Building a teleoperation and demonstration-collection pipeline (LeRobot, spacemouse interface) for a randomized-pose pick-and-place task on a Franka Panda in MuJoCo.
- Training a Diffusion Policy / ACT model on collected demonstrations, with evaluation against a scripted baseline controller.
- Extending the task to peg-in-hole insertion in simulation to connect imitation-learned manipulation with prior 6D pose-estimation and slot-localization work.

LANet: Lightweight Semantic Segmentation for Automated Facade Installation

- Designed a single-stream segmentation network for real-time edge deployment on construction robotics platforms, using multi-scale linear attention and channel attention modules to handle severe class imbalance across frame, glass, and background classes.
- Optimized the architecture to match heavier baselines in accuracy at a fraction of the compute footprint, enabling on-site edge inference without a dedicated GPU server.

Markerless 6D Pose Estimation System

- Designed a 3-stage monocular pipeline (segmentation $\rightarrow$ PnP $\rightarrow$ twist-space residual MLP) for markerless 6D pose estimation of assembly slots.
- Reduced translation error to 12.2 mm ($6.75\times$ improvement) and rotation error to 1.03° ($14.31\times$ improvement) versus standard PnP baselines.
- Achieved sub-100 ms end-to-end GPU inference, with residual correction adding < 0.1 ms ($\leq 0.55\%$) overhead.
- Curated multi-environment datasets of 9,191 segmentation images and 10,000 pose-error regression samples.

Safety-Aware RL for Variable Admittance Control in Robotic CWM Installation

- Formulated contact-rich facade installation as a variable admittance control problem, learning 6D Cartesian stiffness gains online via Soft Actor-Critic (SAC) to regulate contact force in real time.
- Embedded passivity-inspired energy constraints and force-limiting supervisory overrides directly into the low-level controller to guarantee stability and prevent component damage.
- Benchmarked SAC against PPO, TD3, and DDPG in MuJoCo under staged force-noise and pose-uncertainty conditions, then validated the trained policy on a UR5 testbed with a Robotiq FT 300 sensor via zero-shot sim-to-real transfer.
- Deployed the control stack as a ROS2 node running in a Docker container, interfacing the learned policy with the UR5 driver and F/T sensor in real time.